We are reviewing Australia's policies for testing tomato and capsicum seed. As part of this process, we are making changes to existing testing requirements to reduce the risk of infected seed entering Australia. Phase 1 of this review focused on Tomato brown rugose fruit virus (ToBRFV) and Tomato mottle mosaic virus (ToMMV). We are now undertaking Phase 2, which reviews the testing requirements for Pepino mosaic virus (PePMV) and viroids.

Phase 1 (Completed)

Following the initially proposed changes and feedback received from stakeholders, we have updated our import conditions. The key changes include:

- laboratory authorisation - tomato and capsicum seed lots must be tested for ToBRFV and ToMMV by a department authorised laboratory
- updated diagnostic protocols (qPCR) for testing ToBRFV and ToMMV
- a defined threshold cycle/quantification cycle (Ct/Cq) cut-off value of 35 to determine the results of each of the qPCR tests
- updated information required on laboratory reports.

The new requirements became mandatory import conditions on 12 November 2025.

Phase 2

We announced the commencement of Phase 2, a wider review of diagnostic testing requirements for tomato and capsicum seed, on 12 May 2026.

Frequently asked questions – imported seed disease testing requirements

- Prior to 12 November 2025, we allowed all laboratories to conduct ToBRFV and ToMMV testing of tomato and capsicum seed lots. Under the new conditions, only laboratories authorised by us are permitted to perform these tests.
- Laboratories that are not yet authorised must apply for authorisation before testing tomato or capsicum seed for export to Australia.
- The list of authorised laboratories is available on our Department authorised seed testing laboratories webpage.
    - The list is also linked to the applicable BICON cases import permit conditions for tomato and capsicum seed for sowing.
    - The import conditions specify that, from 12 November 2025, only laboratories on this list are permitted to test tomato and capsicum seeds for ToBRFV and ToMMV
- Authorised laboratories are required to keep records of protocols and standard operating procedures, test results, and any decisions about potentially anomalous results.

- Consignments that arrive on or after 12 November 2025 that did not meet the new requirements were offered onshore testing at an approved laboratory in Australia (at the importer's expense),or export or disposal.

- In response to the detection of ToBRFV in Australia, we undertook a reviewing of our seed testing requirements for ToBRFV and ToMMV.
- The updated testing requirements aim to ensure that the risks presented by these viruses remain at an acceptable level, consistent with Australia's appropriate level of protection (ALOP).
- The updates also support improved assurance in offshore laboratory performance and enhance the diagnostic protocols used for seed testing.
- Changes to import conditions were communicated to industry via Import Industry Advice Notices and BICON alerts and updates to the Biosecurity Import Conditions (BICON) system.

- Laboratories that intend to test tomato and capsicum seed for export to Australia must register to be an authorised testing laboratory by completing a registration form confirming they will implement the revised testing requirements.

The department published the list of authorised laboratories in advance of the new testing requirements commencing on 12 November 2025.

Effective 12 November 2025, the following changes have been introduced to the testing protocols for tomato and capsicum seed:

- Specific qPCR protocols are approved for testing for ToBRFV and ToMMV.
- Authorised laboratories are required to use 2 of 3 approved qPCRs for each virus.
- A Ct (Cq) cut-off of 35 must be used to determine the results of each of the qPCR tests.
- The Alkowni et al (2019) and Levitzky et al (2019) end-point PCR tests are no longer approved for testing and have been removed from the list of approved protocols.
  - These 2 end-point PCRs are no longer approved because they are less sensitive than the qPCRs and are not sufficiently sensitive to achieve the appropriate level of protection for Australia.
- Rules for retesting subsamples after potentially anomalous amplifications.
  - Laboratories may retest a subsample seed-homogenate (seed slurry) or RNA extract following certain requirements.
  - Only the approved qPCRs may be used for this retesting.

Most of Australia's requirements for tomato and capsicum seed importation remain unchanged. In particular:

- authorised seed testing laboratories will continue to determine the optimal reagents, cycle times and temperatures for each qPCR protocol

- no changes have been made relating to sampling, sample sizes, and nucleic acid extraction methods.

- The list of approved protocols and requirements for detecting ToBRFV and ToMMV are outlined at our Pathogen testing requirements webpage

- A common Ct cut-off value of 35 has been adopted to improve detection reliability and sensitivity, and to improve consistency between laboratories.
- Applying a common Ct threshold supports consistent result interpretation across different laboratories, leading to reliable regulatory decisions.
- The specified Ct cut-off of 35 balances the likelihood of detecting low levels of virus, while limiting the likelihood of false positives.
- Information from several sources informed this decision, including:
- Australian and international validation studies
- Australian laboratory experiences
- data and other information received from laboratories in response to the draft guidelines.

Establishing a list of authorised laboratories has a number of benefits, including:

- Improved assurance of laboratory performance through the authorisation process
  - Authorised laboratories are aware of the requirements and their responsibilities
  - Test results from laboratories that have not been authorised will not be accepted as evidence to meet Australia's biosecurity import conditions
- Targeted and consistent communication
  - By establishing a list of authorised laboratories, communication with laboratories improves by ensuring relevant messages reach the relevant laboratories
  - We are better able to consider specific concerns of laboratories
- More effective management of noncompliance
  - Having a list of authorised laboratories allows us to remove laboratories that either are noncompliant or simply no longer interested in testing for the Australian market, with minimal ongoing engagement required
- Improved assurance of record keeping and documentation, aiding traceability
  - For laboratories to be authorised, they are required to retain records of all testing performed for seed intended for export to Australia
  - If potentially infected seed lots need to be traced, the retention of test results and other documentation is essential.

There are 2 main reasons why the department requires 2 qPCR protocols for each virus:

- Minimising false negatives due to viral variations.
  - Variations in viral genomic regions targeted by qPCR can disrupt primer and probe binding and compromise the test, potentially leading to false negatives. Virus variants that have not been recorded before may arise in the future and some existing variants will not have been isolated and sequenced.

    - Employing 2 qPCR assays with distinct primer and probe sets significantly reduces the likelihood of detection failure due to genetic variations
- Enhancing diagnostic coverage by targeting different regions of the viral genome.
    - The pairs of qPCRs selected for each virus are more likely to detect all the virus variants than any one of the qPCRs would if used on its own.
    - The CaTa28 qPCR targets a region of the ToBRFV genome that differs from the region targeted by the CSP1325 and qs1/p1/qas2 qPCRs. Similarly, the CaTa9 qPCR targets a region of the ToMMV genome that differs from the region targeted by the CSP1572 and ToMMV2 qPCRs.

- Yes. We will accept test results from multiplex qPCRs provided that the laboratory has verified that the multiplex is as sensitive as the most sensitive of the 2 singleplex qPCRs and that each qPCR is working in the multiplex.
- The sensitivity and repeatability of the multiplex must be verified against the singleplex format.

- We may consider alternative proposals on a case-by-case basis, such as for the recognition of other qPCR primers and protocols, following the current review.
- Testing laboratories seeking approval to use alternative protocols to those approved must submit a detailed proposal to us that includes a validation report.
- Please send requests via email to imports@aff.gov.au (please title the subject line of the email 'Plant T2 - Seed for sowing').
- We will only consider proposals when resources allow.

- PCR tests must be conducted on subsamples of no more than 400 seeds per test (i.e. 50 tests, each on 400 seeds). When the total sample size is 20,000 seeds, this is typically implemented as 50 separate tests, each analysing 400 seeds.
- A larger subsample size may risk false negative test results. Increasing the subsample size may lead to reduced sensitivity of the PCR test, thereby increasing the risk of false negatives.
- While the 400-seed limit is maintained as a general requirement, we are open to considering proposals for increased subsample size on a case-by-case basis following the current review.
- Testing laboratories seeking approval to use larger subsample sizes must submit a detailed proposal to us. Please submit requests via email to imports@aff.gov.au (please title the subject line of the email 'Plant T2 - Seed for sowing').
- We will only consider proposals when resources allow.

- Only certain requirements for capsicum and tomato seed testing are being updated at this time.
    - These changes apply specifically to the detection of ToBRFV and ToMMV
    - All other testing requirements remain unchanged.
- Australia's requirements for testing capsicum and tomato seed that remain unchanged include:
    - Requirements for sample sizes and subsample sizes
    - Requirements for reporting test results.
- Australia currently does not prescribe specific requirements for:
    - RNA extraction methods for capsicum and tomato seeds
    - qPCR cycle conditions (e.g. times and temperatures)

    - Storage conditions for RNA extracts or seed homogenates.

- No. All laboratories that intend to test tomato and capsicum seed for export to Australia must submit a laboratory registration form, regardless of existing accreditation.
- We are communicating with every laboratory that applies to become authorised to understand their specific testing conditions.

- Yes. The list of authorised laboratories was published on our website on 1 October 2025, that provided a 6-week transition period before the import conditions became mandatory on 12 November 2025.
- Consignments that arrived on or after 12 November 2025 that did not meet the new requirements were offered onshore testing at an approved laboratory in Australia (at the importer's expense), export or disposal.

- Seeds that have been imported into Australia and released from biosecurity control prior to 12 November 2025 are not affected by these changes.

- Import permits were varied by the department on 12 November 2025 to reflect revised import conditions.

- Yes. The pathway for testing tomato and capsicum seed at the Naktuinbouw laboratory will remain approved.

- We announced the proposed changes via an Import Industry Advice Notice and BICON alert on 17 April 2025, and invited feedback from stakeholders.
- We carefully considered all feedback received and, where requested, held meetings with stakeholders to further discuss the proposed changes.
- On 3 September 2025, we issued a follow-up Industry Advice Notice and BICON alert to announce the final policy. We also notified stakeholders directly via email.
- On 16 September 2025 , we held 2 online webinars to explain the new policy and address stakeholder questions. Recordings of these sessions are available above.

Background

Emergency measures for tomato and capsicum seed

- The department implemented emergency measures for ToBRFV and ToMMV in March 2019 to address the increasing global distribution and threat to the Australian tomato and capsicum production industries.
    - ToBRFV naturally infects tomato and capsicum, and ToMMV naturally infects tomato, resulting in unmarketable fruit.
    - ToMMV infection has been observed to break resistance in cultivars resistant to the related Tomato mosaic virus.

    - Potentially at-risk Australian industries include tomato, capsicum and chilli with values of annual production of $570.6, $215.8 and $13.5 million, respectively during 2022/23 (total $799.9 million).
    - Like other Tobamoviruses, there is no effective seed treatment option available for commercial quantities of ToBRFV or ToMMV infected seed.
    - ToMMV is NOT known to be present in Australia.
- The new seed testing requirements that have been introduced on 12 November 2025 are an update to these existing emergency measures.

- The SPS Agreement permits Australia to implement emergency actions, including emergency measures, when a new or unexpected phytosanitary risk is identified.
- Ongoing need for the continuance of measures will be evaluated by a pest risk analysis as soon as possible, to ensure that the measures are technically justified.

- Yes, pooling - the process of combining 2 or more individual samples to form a single sample, which is then analysed with a diagnostic test - is permitted for small seed lots only.
- Small seed lots that are being pooled for testing must be pooled by the testing laboratory.
- If pooled seed lots fail testing, the individual seed lots from the pooled sample may be retested to determine the infected lot/s. This will require resampling 20% of each individual seed lot for testing. If the infected lot/s cannot be identified, the department will take action based on the initial pooled test results. i.e. all seed lots that made up the original pooled samples will not be permitted entry into Australia.

Tomato brown rugose fruit virus (ToBRFV)

- ToBRFV is a member of the Tobamovirus genus:
    - It is transmitted through propagation materials (seeds, plants for planting, grafts, cuttings), and spreads locally by contact (direct plant to plant contact, contaminated tools, hands, or clothing and by bees)
    - It can remain infective in plant debris and contaminated soil for months
    - It has plant disease resistance breaking capabilities for commercial cultivars of tomato
- ToBRFV naturally infects tomato and capsicum fruit resulting in unmarketable fruit:
    - On tomato, foliar symptoms include chlorosis, mosaic and mottling with occasional leaf narrowing
    - Necrotic spots may appear on peduncles, calyces and petioles
    - Fruit show yellow or brown spots, with rugose symptoms rendering the fruits non-marketable
    - Fruits may be deformed and have irregular maturation
    - On capsicum, foliar symptoms include deformation, yellowing and mosaic. Capsicum fruits are deformed, with yellow or brown areas or green stripes.

- There is no risk to food safety or human health from eating tomatoes or capsicums with this disease.

- ToBRFV in tomatoes was first reported in southern Israel in 2014 and in Jordan in 2015. Subsequently, ToBRFV was later reported from more than 40 countries all over the world (across continents Africa, America, Asia and Europe).
- However, given the global nature of the seed production cycle, it is likely that the distribution of the virus may be greater than that reported in published records.

- The department implemented emergency measures for ToBRFV in March 2019 to address the increasing global distribution and threat to the Australian tomato and capsicum production industries.
- ToBRFV was detected for the first time in Australia in August 2024 in the Northern Adelaide Plains, South Australia.
- In January 2025 the virus was also detected at a single property in Victoria linked to a direct movement of tomato seedlings from South Australia.
- On 29 May 2025, the National Management Group (NMG) determined that it is no longer technically feasible to eradicate ToBRFV from Australia. For more information on the decision visit National Management Group Communiqué: Tomato brown rugose fruit virus – 29 May 2025 - DAFF.
- The pest and regulatory status of ToBRFV in Australia is currently under review. For further information on the outbreak visit outbreak.gov.au and pir.sa.gov.au/tobrfv.

- ToBRFV has rapidly expanded its global distribution since it was first reported in 2014. The first outbreak of this virus was reported on tomatoes in Israel in 2014, and it has since been reported in Europe, the Middle East, China, Mexico, the USA and Australia.
- Global movement via seed appears to be the only credible explanation for the observed intercontinental movement of ToBRFV.
- ToBRFV is a member of the Tobamovirus genus. These viruses are known to be seed-borne, can remain viable in seeds for months, and are known to be associated with the seed coat and endosperm.
- Several tobamoviruses are recognised as economically important seed-borne quarantine pests, including Cucumber green mottle mosaic virus (CGMMV), Kyuri green mottle mosaic virus (KGMMV) and Zucchini green mottle mosaic virus (ZGMMV).
- The European and Mediterranean Plant Protection Organization (EPPO) advised in a recent alert that they consider ToBRFV to be seed-borne.
- Mexico has introduced phytosanitary measures for the testing of imported seed for sowing of host species and allege extensive positive test results on imported seeds originating from 14 countries across 3 continents, including Asia.
- Turkey implemented emergency measures in 2019, for the RT-PCR testing of imported tomato and capsicum seeds originating from Jordan, Germany, Israel, Italy and Mexico.

- New Zealand implemented emergency measures for imported tomato and capsicum seeds, in 2019. Measures are that seeds be sourced from pest free areas or a 'pest-free place of production' or tested by ELISA and found free of ToBRFV.

- The emergency measures apply to imported tomato (Solanum lycopersicum) and capsicum (Capsicum annuum species complex) seed for sowing.
- The emergency measures on capsicum are applied to Capsicum annuum species complex (C. annuum, C. chinense and C. frutescens) because:
    - ToBRFV is likely to be seed-borne in all 3 species, which are closely related
    - hybrids of the three species are commonly bred and distributed globally
    - seeds of the 3 species within the complex are similar in appearance and thus difficult to distinguish
    - seed traders usually only identify seeds using the common name ‘capsicum’.

Tomato mottle mosaic virus (ToMMV)

- ToMMV is a member of the Tobamovirus genus:
    - It is transmitted through propagation materials (seeds, plants for planting, grafts, cuttings), and spreads locally by contact including direct plant-to-plant contact, contaminated tools, hands, or clothing and by bees
    - Tobamoviruses can remain infective in seeds, plant debris and contaminated soil for months
    - It has plant disease resistance breaking capabilities for commercial cultivars of tomato.
- ToMMV naturally infects tomato and capsicum resulting in unmarketable fruit:
    - On tomato, symptoms include severe necrotic lesions, chlorosis, leaf distortion, mottle and systemic crinkling symptoms and fruit necrosis. Disease incidences of up to 87% have been reported
    - On capsicum, symptoms include foliar mottle, shrinking and necrosis
    - On eggplant, disease incidences of 20-40% have been reported. However, the symptoms observed (dark purple mottle on flowers, and mosaic and distortion on leaves) on eggplant were the result of a mixed infection involving ToMMV and TMGMV (Tobacco mild green mosaic virus) and symptoms caused by ToMMV alone on eggplant have not been reported.
    - Chickpea has been reported as a natural host, but infections appear to be asymptomatic.

- At the time emergency measures were introduced by the department in 2019, ToMMV was known to be present in Brazil, Mexico, USA, China, Iran, Israel and Spain. Since then, ToMMV has been reported from Mauritius, India, Japan, Czech Republic, Germany, France and the Netherlands.
- However, given the global nature of the seed production cycle, it is likely that the distribution of the virus may be greater than that reported in published records.

- The emergency measures apply to imported tomato (Solanum lycopersicum) and capsicum (Capsicum annuum species complex) seeds for sowing.
- The emergency measures were applied on capsicum and tomato seeds because of detections of ToMMV on imported capsicum and tomato seeds for sowing, confirming ToMMV is seed-borne.

- The emergency measures on capsicum are applied to Capsicum annuum species complex (C. annuum, C. chinense and C. frutescens) because:
    - ToMMV is likely to be seed-borne in all 3 species, which are closely related
    - Hybrids of the 3 species are commonly bred and distributed globally
    - Seeds of the 3 species within the complex are similar in appearance and thus difficult to distinguish
    - Seed traders usually only identify seeds using the common name ‘capsicum’.

- Growers are strongly encouraged to familiarise themselves with the symptoms of ToMMV infection on host crops and ensure that they have effective on-farm biosecurity practices in place, especially being vigilant for any unusual signs of virus infection on host crops.
- Growers are also advised to contact their seed suppliers to seek their assurance about what specific testing has been applied to batches of seeds and whether they have been appropriately tested for ToMMV.
- Reported symptoms of ToMMV infection include:
    - On tomato, symptoms include severe necrotic lesions on both leaves and fruit, chlorosis and deformation of leaves, mottle and systemic crinkling symptoms and fruit necrosis
    - On capsicum, symptoms include foliar mottle, shrinking and necrosis.
- Key Actions:
    - Report immediately: Contact the Exotic Plant Pest Hotline at 1800 084 881 or through the DAFF website
    - Check plants regularly: Look for unusual signs of disease, including those caused by ToBRFV or ToMMV
    - Source seeds carefully: Purchase local seeds, seedlings, and graft material from reliable suppliers that are free of the virus
    - Follow biosecurity advice: Implement measures like disinfecting tools and removing weeds that can host the virus.
- Report immediately: Contact the Exotic Plant Pest Hotline at 1800 084 881 or through the

Source: https://www.agriculture.gov.au/biosecurity-trade/import/goods/plant-products/seeds-for-sowing/emergency-measures-tomato-and-capsicum-seed